

## Addition and subtraction of algebraic terms



1

a  $4x + 4y$

c  $3x + 2y$

e  $21xy$

g  $20p^2 - 11p$

i  $3x + 19y$

k  $7x^2 + x$

m  $25y - 9y^2$

o  $3ab + 2c$

q  $25x + 2y$

s  $5m + 12n$

b  $15m + 19n$

d  $25m + 8n$

f  $8q$

h  $3n^2 - 4m + 2m^2$

j  $18a - 8b$

l  $8x - 7y + 4z$

n  $11a - 7b + 4c$

p  $23j + 2k$

r  $12y$

t  $6s - 2t$

2

a  $-3x - y + 10z$

c  $12x^2 + 9x + 5$

e  $11mn - 2m$

g  $st + 7s + 7t$

i  $8x + 6y$

k  $2pq + 11p^2q$

m  $20xyz + 23xy$

o  $-6rk + 3r - 6k$

q  $-5n + 7$

b  $-6xy + 2yz + 6xz$

d  $-16x + 10y + 41$

f  $24x^2 + 5x - 15$

h  $19ab + 3bc - 9b$

j  $7p + 7$

l  $17x^2$

n  $-12x - 4y$

p  $-14vw - 2v^2w$

r  $-2n + 5$

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## Multiplication of algebraic terms

3

a  $60u$

b  $-21u$

c  $10x$

d  $33y$

e  $-72x$

f  $24u$

g  $-6w$

h  $-30y$

4

a  $x^2$

b  $6x^2$

c  $4y^2$

d  $15a^2$

e  $16x^2y$

f  $27b^2$

g  $100a^2$

h  $16x^2$

5

a  $72mn$

b  $24wz$

c  $50rs$

d  $54rs$

e  $42u^2v^8$

f  $224p^3q^3$

g  $8ab$

h  $-21wz$

i  $20rs$

j  $-60r^8s^7$

k  $-12p^5q^5$

l  $-10a^4b^3$

6

a  $96rs$

b  $630wxy$

c  $-80wxy$

d  $-480hkrs$

e  $-90xy$

f  $-80a^3$

g  $90x^2y$

h  $-40hijk$



## Division of algebraic terms

7

a  $x$

b  $3v$

c  $\frac{m}{4}$

d  $\frac{1}{4}$

e  $xy$

f  $7q$

g  $\frac{4n}{5}$

h  $\frac{6}{w}$

i  $\frac{1}{q}$

j  $10v^4$

k  $-6a$

l  $-11$

m  $-\frac{1}{11}$

n  $-4$

o  $\frac{4}{9}$

p  $-ac$

q  $-\frac{1}{j}$

r  $-\frac{6}{k}$

s  $-\frac{1}{b}$

t  $-\frac{2}{3}$

u  $-3w^5$

v  $12$

8

a  $\frac{m}{8}$

b  $5$

c  $2$

d  $-4$

e  $-11s$

f  $6$

g  $2n$

h  $3x$

i  $3r$

j  $-2$

k  $11x$

l  $10c$

9

a  $6$

b  $2$

c  $3x^2$

d  $14$

e  $3a$

f  $\frac{9}{2}$

g  $18x$

h  $-4x$

i  $33xy$

j  $15y^2$

k  $10x$

l  $y$

## Applications

10  $(10 + a)$  cm

11

a 24



b  $6y$

12

a  $3x$  cm

b The water level is increasing by the same amount each hour.

13  $5p + 10q$

14

a  $(300 + 70t)$  m

b Uther is paddling at a constant pace over the first  $t$  minutes.

15

a  $9 + y$

b It represents that it takes Sally 9 minutes longer than the bus trip to get to school.

16

a  $44WL$  cm<sup>3</sup>

b  $132WLH$  cm<sup>3</sup>

c  $3H$

d 264

17

a  $6xy$  cm<sup>3</sup>

b  $48xyz$  cm<sup>3</sup>

c  $8z$

d 120